

Bioinformatics assignment

Code Alpha

Task no.1: DNA/Protein Sequence Analysis

Step1:

UniProtKB/Swiss-Prot: P68871.2

[GenPept](#) [Identical Proteins](#) [Graphics](#)

```
>sp|P68871.2|HBB_HUMAN RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-globin;
AltName: Full=Hemoglobin beta chain; Contains: RecName: Full=LVV-hemorphin-7; Contains:
RecName: Full=Spinorphin
MVHLTPEEKSAVTALWGKVNVDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDAMGNPKVKAHGKKVLG
AFSDGLAHLDNLKGTATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVAN
ALAHKYH
```

Step 2:

i Your search is limited to records **exclude: non-redundant RefSeq proteins**

Job Title	sp P68871.2		
RID	3CJVA7WS014	Search expires on 06-21 16:12 pm	Download All 
Program	BLASTP 	Citation 	
Database	ClusteredNR	See details 	
Query ID	P68871.2		
Description	RecName: Full=Hemoglobin subunit beta; AltName: Full=Beta-		
Molecule type	amino acid		
Query Length	147		
Other reports	Distance tree of results	Multiple alignment	MSA viewer 

☒	1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	263	263	87%	9e-88	100.00%	175	AAA35952.1
☒	1 member(s), 1 organism(s)	human	beta-globin [Homo sapiens]	277	277	100%	2e-93	93.20%	147	AAA35597.1
☒	1 member(s), 1 organism(s)	giant panda	hypothetical protein PANDA_015769 [Ailuropoda melanoleu...	275	446	100%	3e-91	90.48%	258	EFB18430.1
☒	1 member(s), 1 organism(s)	rabbit	hemoglobin subunit beta-1/2 isoform 1x [Oryctolagus cunicu...	253	253	100%	8e-84	90.48%	170	NP_001300972.1
☒	7 member(s), 6 organism(s)	dog_coyote_wolf_fox	hemoglobin subunit beta-like [Canis lupus familiaris]	275	275	100%	1e-92	89.80%	147	NP_001257812.1
☒	185 member(s), 76 organism(s)	placentals	hemoglobin subunit delta [Ailuropoda melanoleuca]	273	273	100%	6e-92	89.80%	147	NP_001291814.1
☒	19 member(s), 18 organism(s)	placentals	hemoglobin subunit beta-like [Ailuropoda melanoleuca]	273	273	100%	7e-92	89.80%	147	NP_001291812.1
☒	5 member(s), 5 organism(s)	Old World monkeys	PREDICTED: LOW QUALITY PROTEIN: hemoglobin subun...	270	270	100%	8e-91	89.80%	147	XP_011830554.1
☒	9 member(s), 6 organism(s)	bats	hemoglobin subunit beta [Rhynchonycteris naso]	271	271	100%	2e-91	89.12%	147	KAM8815943.1
☒	1 member(s), 1 organism(s)	black flying fox	Hemoglobin subunit beta [Pteropus alecto]	272	518	100%	2e-88	89.12%	363	ELK09417.1

Analysis:

The table documents several key homologous sequences across different mammalian groups:

- Human (*Homo sapiens*): The top hits are for beta-globin, with one sequence showing 100% identity over 100% coverage, confirming a perfect match to the human reference.
- Giant Panda (*Ailuropoda melanoleuca*): Shows high homology with a 90.48% identity and an E-value of $3e^{-91}$, suggesting a very conserved hypothetical protein related to hemoglobin.
- Rabbit (*Oryctolagus cuniculus*): A sequence for hemoglobin subunit beta-1/2 also shows 90.48% identity, further illustrating the conservation of this protein in mammals.
- Other Species: Significant matches are also found in dogs, wolves, foxes, Old World monkeys, bats, and black flying foxes, with identities consistently around the 89% to 90% range.

Explanation of all the core metrics in table:

1. Query Coverage: This indicates how much of your input sequence is covered by the match. Most results in the table show 100% Query Cover, meaning the entire query sequence was successfully aligned with the subject sequence in the database.
2. E-value (Expect Value): This is the most critical statistical metric. Values in the table such as $2e^{-93}$ or $1e^{-92}$ are essentially zero, which indicates that the matches are highly significant and extremely unlikely to have occurred by chance.
3. Per. Ident (Percentage Identity): This shows how many amino acids are identical between the two sequences. High percentages (ranging from 89.12% to 100%) confirm that these sequences are closely related homologs.

Task 2: Multiple sequence alignment:

I took the sequence of protein showing high percentage identity with the my query sequence in FASTA formats. Perform the multiple sequence alignment using clustal

omega. The sequence is from *Homo sapiens*, *Ailuropoda melanoleuca*, *Oryctolagus cuniculus*, *Mus musculus*, and *Pan troglodytes*.

AAA35952.1	-----	0
EFB18430.1	MVHFTAEEKAAVVS L WARVNV L VGG E VLGRLLVVPWTQ R FF D SFGNLSSES A IMGNPK	60
NP_001129304.1	-----	0
NP_001300972.1	-----	0
NP_058652.1	-----	0
AAA35952.1	-----MVHLTPEEK	9
EFB18430.1	VKAHGK K VLT S FGNAV K HMD L KDTFAELSELHCD K LHVDPENFKQ P RDTMVHLTGEEK	120
NP_001129304.1	-----MVHFTAEEK	9
NP_001300972.1	-----MVHLSSEK	9
NP_058652.1	-----MVHLTDAEK	9
	***: : **	
AAA35952.1	SAVTALWGK V NDEVGGEALGRLLVVPWTQ R FFESFGDLSTPD A VMGNPKVKAHGK K VL	69
EFB18430.1	AAVTGLWSK V NDEVGGEALGRLLVVPWTQ R FFDSFGDLSTPD A VMNPNK V KAHGK K VL	180
NP_001129304.1	AAVTSLSK M NVEEAGGEALGRLLVVPWTQ R FFDSFGNLS S PSAILGNPK V KAHGK K VL	69
NP_001300972.1	SAVTALWGK V NVEEAGGEALGRLLVVPWTQ R FFESFGDLSSAHVMSNPK V KAHGK K VL	69
NP_058652.1	SAVSCLWAK V NPDEVGGEALGRLLVVPWTQ R YFDSFGDLSSASAIMGNPK V KAHGK K VI	69
	:** : **.*:* :*,*****:*;***:** :*:.*****:	
AAA35952.1	GAFSDGLAHL D NLKGTFATLSELHCD K LHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQ P	129
EFB18430.1	NSFSEGLKNL D NLKGTFAKLSELHCD K LHVDPENFKLLGNVLVCVLAHHFGKEFTPQVQ A	240
NP_001129304.1	TSFGDAIK N MDNLKPAFAKLSELHCD K LHVDPENFKLLGNVMVILATHFGKEFTPEVQ A	129
NP_001300972.1	AAFSEGLNHL D NLKGTFAKLSELHCD K LHVDPENFRLLGNLVVVL S HHFGKEFTPQVQ A	129
NP_058652.1	TAFNEGLKNL D NLKGTFASLSELHCD K LHVDPENFRLLGNAIVIVLGHHLGKDFTPAAQ A	129
	:*:. : :***** :*,*****:*****.* :*,*:**:***.*	
AAA35952.1	HLSVVWLMPWPTT-PGQ R AGLCAGP-SLWQIIHPTSAGCLSESGGWCG	175
EFB18430.1	AYQKV V AGVANALAHKYH-----	258
NP_001129304.1	AWQKL V SAVAIALAHKYH-----	147
NP_001300972.1	AYQKV V AGVANALAHKYHXDLFPSAK N YGDIMKPLEHL---TSG----	170
NP_058652.1	AFQKV V AGVATALAHKYH-----	147
	. : : :	

The multiple sequence alignment (MSA) results in the source show a comparison between three protein sequences: AAA35952.1, EFB18430.1, and NP_001300972.1, NP_001300972.1, and NP_058652.1. This alignment identifies conserved evolutionary patterns and structural similarities across these protein variants, which are characteristic of the hemoglobin beta (HBB) family discussed in our previous conversation.

Interpretation of Alignment Symbols

The consensus line at the bottom of each alignment block uses standard bioinformatics symbols to denote the level of conservation:

Asterisk (*): Indicates positions where all three sequences have an identical amino acid, representing high evolutionary conservation.

Colon (:): Indicates positions where the amino acids have strongly similar biochemical properties.

Period (.): Indicates positions where the amino acids have weakly similar properties.

Blank Space: Represents a position with no conservation, where the sequences differ significantly.

Explanation of results:

The alignment confirms that these three sequences share a common ancestral domain related to the hemoglobin beta chain but differ in their terminal extensions. The high degree of central conservation indicates that the core structural and functional properties, specifically the globin fold, are maintained across these different variants, despite the variations observed at their start and end points.

Task 3: Protein structure prediction

FASTA ▾

beta-globin [Homo sapiens]

GenBank: AAA35597.1

[GenPept](#) [Identical Proteins](#) [Graphics](#)

>AAA35597.1 beta-globin [Homo sapiens]
MVHXTPEKSAVTALWGKVNVDXGGEALGXLLVVYPWTQRFFESFGDLSTPDVAMGNPKVKAHGKKVLGAFSDGLAHLNLTGKTFATLSLHCDKLLHVDPENFR 105
AFSDGLAHLNLTGKTFATLSLHCDKLLHVDPENFRLLGNVXVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH 147
ALAHKYH

Start a New Modelling Project ⓘ

Target Sequence(s):  Target MVHLPTEKSAVTALWGKVNVDXGGEALGXLLVVYPWTQRFFESFGDLSTPDVAMGNPKVKAHGKKVLGAFSDGLAHLNLTGKTFATLSLHCDKLLHVDPENFR 105
(Format must be FASTA, Clustal, plain string, or a valid UniProtKB AC)
Target LLGNVLCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH 147

Add Hetero Target

 Reset

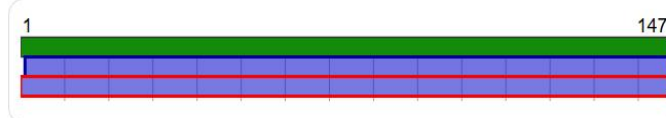
Project Title:

Email:

Search For Templates

Build Model

By using the SWISS-MODEL server, you agree to comply with the following [terms of use](#) and to cite the corresponding [articles](#).



Model 02

Structure Assessment

Compare ☐

Download files Display files

Oligo-State
Monomer

GMQE
0.94

QMEANDisCo Global:
0.86 ± 0.07

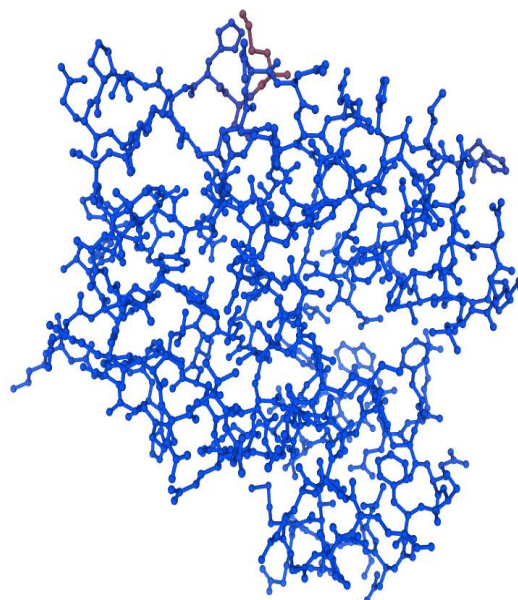
Ligands
1 x HEM

QMEANDisCo Local

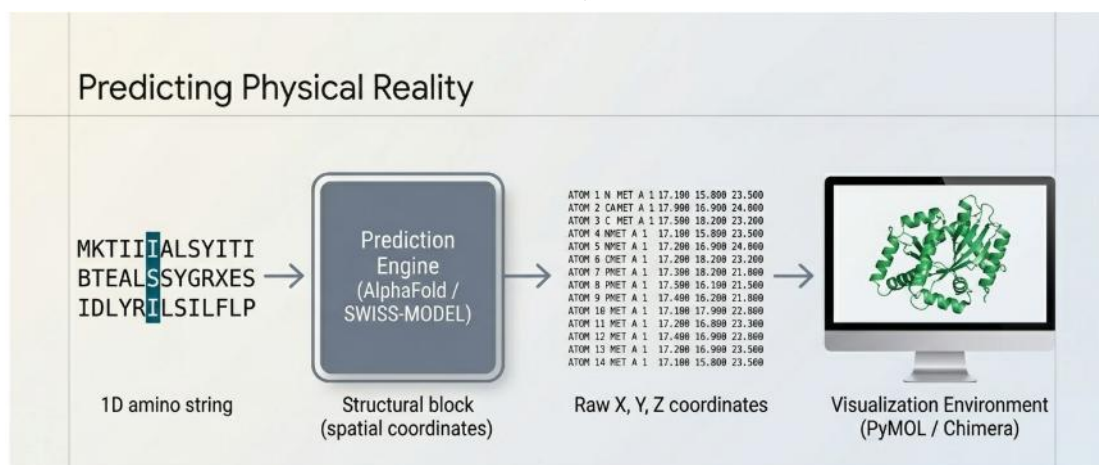
QMEAN Z-Scores

Template
6kye.1.D Hemoglobin subunit beta
The crystal structure of recombinant human adult hemoglobin

Seq Identity
100.00%
Coverage



BallAndStick



Summary of assignment and bioinformatics tool analysis:

The Bioinformatics Toolbox

Tool / Database	Pipeline Phase	Input Type	Primary Output	Core Purpose
NCBI BLAST	Phase 1	Single Sequence	Homology Matches	Sequence Similarity Search
Clustal Omega / MUSCLE	Phase 2	Multiple Sequences	Alignment Grid	Evolutionary Conservation
SWISS-MODEL / AlphaFold DB	Phase 3	Single Sequence	3D Coordinate Map	Structure Prediction
PyMOL / Chimera	Phase 3	3D Coordinate Map	3D Visual Rendering	Structure Visualization